

**Econ 630, Spring 2012:
Final Exam**

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Name: _____

The exam consists of 3 questions. Your answer should be clearly labeled and legible. Show all your work. No credit will be given for answers that are not justified.

1. (20 pts) Consider a seller who wants to auction two identical objects. There are three bidders ($i = 1, 2, 3$) participating in the auction and each bidder has an identical and commonly known valuation for the object of v for the first unit and 0 for the second unit. Assume that the seller's valuation is normalized to 0. The buyer conducts an all-pay auction with 0 reserve price and no participation fee. The two highest bidders win one unit, but all bidders pay their bids independent of whether they have won the object. Let $F(b)$ denote a symmetric Nash equilibrium CDF of bids in this game.
- (a) (3) From the point of view of bidder i , what is the probability that he loses the auction given that the other two bidders bid according to $F(b)$?
- (b) (3) From the point of view of bidder i , what is the probability that he wins one of the objects given that the other two bidders bid according to $F(b)$?
- (c) (4) Write the expected payoff of bidder i , $\pi_i(b_i)$ given that the other two bidders bid according to $F(b)$?

(d) (10) Derive the Nash equilibrium CDF $F(b)$ with its corresponding support.

2. (40 pts) Consider the following dynamic game of complete and perfect information. A buyer wants to purchase one unit of a good valued by her at $v > 0$. There are two potential sellers ($i = 1, 2$) of this item whose production cost is normalized to 0. The buyer incurs a small transaction cost $c \in (0, \frac{v}{2})$ from visiting each seller. In period 0, the buyer chooses whether to visit seller 1. If visited, in period 1 seller 1 makes the buyer a price offer of p_1^1 . Then, the buyer chooses whether to 1) accept the offer (*accept* p_1^1), 2) reject the offer and quit her search (*quit*) or 3) reject the offer and visit the second seller (*visit* 2). If visited, in period 2, seller 2 observes p_1^1 and makes a price offer p_2^2 . The buyer, then, chooses whether 1) *accept* p_2^2 , 2) *quit* the search, or 3) *visit* 1 to obtain another price quote p_1^3 . The game ends whenever the buyer purchases the good from one of the sellers or quits her search. All players are perfectly patient (i.e. $\delta = 1$). Suppose that in the case of indifference, the buyer breaks the indifference in favor of purchasing the good. If indifferent between purchasing from a sellers and visiting the competing seller, the buyer flips a fair coin.

(a) (5) Consider the subgame, in which the buyer visits seller i in stage t . From the point of view of seller i , this subgame is equivalent to the subgame, in which the buyer visit seller i in stage $t + 2$. Let (p_i^*, p_j^*) denote seller i 's and j 's SPNE prices, in which the sellers make the same price offer in every equivalent subgame. Given (p_i^*, p_j^*) , specify the buyer's SPNE strategy $d_i^*(p_i^*, p_j^*)$ in the subgame, in which seller i was the last to make a price offer.

(b) (5) Specify seller i 's best response $p_i^*(p_j^*)$ as a function of seller j 's pricing strategy p_j^* .

(c) (5) Find (p_i^*, p_j^*) that corresponds to an SPNE, in which the sellers make the same price offer in every equivalent subgame.

(d) (5) Specify the buyer's SPNE strategy in period zero d_0^* . What is the expected payoff for each player in this SPNE?

(e) (5) Is the SPNE you found Pareto efficient? Why/Why not?

(f) (5) Suppose now that $c = 0$ (the buyer became a sophisticated online shopper). Find the SPNE prices (p_i^{**}, p_j^{**}) , in which the sellers make the same price offer in every equivalent subgame.

(g) (5) Specify the buyer's SPNE strategy in period zero d_0^{**} . What is the expected payoff for each player in the SPNE?

(h) (5) Is the SPNE you found in part g) Pareto efficient. Why/Why not?

3. (40 pts) Consider a game of charitable giving, in which donors care about the esteem that they receive from being considered generous by the public. There are two types of donors, generous (G) and miserly (M), which are equally likely in the population (i.e. prior belief $\alpha = Pr(G) = \frac{1}{2}$). The type of the donor is private information. Each donor has to choose the amount x to donate to a charity. The individual's utility function is given by $u_i(\theta_i, x, q) = -(x - \theta_i)^2 + qE$ where $\theta_i(G) = 1$ and $\theta_i(M) = \frac{1}{2}$. The term qE stands for the expected esteem of the donor, where q denotes the public's equilibrium belief that the donor is generous and E denotes the level of esteem the donor gets from being considered generous. In what follows, let's normalize $E = 1$. Note that q can be different from the prior α if the amount donated by the donor signals the donor's type. The timing of the game is as follows:
- The charity commits to either disclose donations or keep them private, which becomes common knowledge.
 - Each donor chooses how much to donate to the charity.
 - The donations become publicly observable if the charity has committed to disclosure. Otherwise, no information is revealed.
 - The public forms a belief q regarding the type of each donor and donors' payoffs are realized.
- (a) (7) Suppose that the charity adopts a policy of not revealing the donors' donation amount. How much would each type of donor donate to the charity in this case ($x^n(G), x^n(M)$) in a PBE of this game? What should the equilibrium q^n be in this case? What would be the expected individual donation and each types' equilibrium payoff?

- (b) (10) Suppose now that the charity commits to a public disclosure of donations. Find the pooling PBE donation amount $(x^p(G), x^p(M))$ that results in the highest donation amount by each type. Carefully specify the belief structure that supports this equilibrium. Is the charity better off or worse off with disclosure if disclosure results in $(x^p(G), x^p(M))$? How about individual donors?

- (c) (6) The Cho-Kreps intuitive criterion ensures that in equilibrium $Pr(\theta|x) = 0$ for all x that are equilibrium dominated for type θ as long as x is not equilibrium dominated for all types. Recall that x is equilibrium dominated for type θ given $x^p(\theta)$, if $u^p(\theta, x^p, q^p) > u(\theta, x, q)$ for all q . Formulate the Cho-Kreps intuitive criterion for the pooling equilibrium you found in part b).

- (d) (7) Does the PBE specified in part b) satisfy the intuitive criterion? If not, can you specify a belief structure satisfying the intuitive criterion that also supports $(x^p(G), x^p(M))$ as a PBE?

- (e) (10) Consider now a separating PBE that satisfies the intuitive criterion. Find the highest amount that each type will donate $(x^s(G), x^s(M))$ under this equilibrium. Carefully specify the belief structure that supports this equilibrium. What is the expected individual donation in this case. Is the charity better off or worse off with disclosure if disclosure results in $(x^s(G), x^s(M))$? How about individual donors?